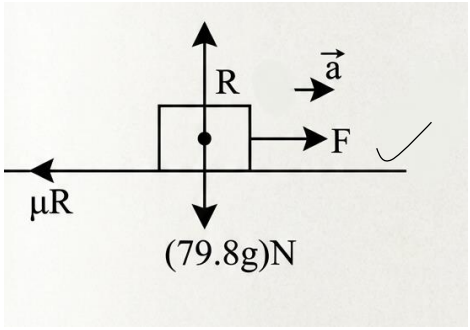
MARKING GUIDE FOR THE ITEMS OF THE SEMINAR HELD AT KISUBI MAPEERA SS -2026

ITEM 1

Item 1(a)

Given: $u = 3.6\ km\ h^{-1} = 3.6 \times 1000 / 3600 = 1\ m\ s^{-1}$



$$P = FV$$
$$180 = F(1)$$
$$F = 180\ N$$
$$R = mg$$
$$R = 79.8 \times 9.8 = 782.838\ N$$

Applying Newton's second law (F − μR = ma):

$$F - \mu R = ma$$
$$180 - (0.25)(782.838) = 79.8\ a$$
$$a = -0.197\ m\ s^{-2}$$

Using $v^2 = u^2 + 2as$:

$$v^2 = 1^2 + 2(-0.197)(2)$$
$$v = 0.4604\ m\ s^{-1}$$

Resultant velocity:

$$V_R = \sqrt{(V_x{}^2 + V_y{}^2)}$$
$$V_x = 12\ cos\ 54° + V = 12\ cos\ 54° + 0.4604 = 7.5138\ m\ s^{-1}$$
$$V_y = 12\ sin\ 54° = 9.7082\ m\ s^{-1}$$

$$V_R = \sqrt{(7.5138^2 + 9.7082^2)} = 12.2763\ m\ s^{-1}$$

Direction: $\alpha = tan^{-1}(V_y/V_x) = tan^{-1}(9.7082 / 7.5138) = 52.26°$

Horizontal range:

$$x = V_R \cdot t \cdot cos\ \alpha$$

$$x = 12.2763(5)\ cos\ 52.26° = 37.57\ m$$

Since $x < 50\ m$, the ball will NOT enter the goal.

Origin of Friction:
The actual area of contact between the player's shoes and the surface is very small, and this creates a high pressure at contact points. This pressure pushes the contact points to form cold welds. These welds have to be broken for motion to occur, which causes friction.

The Spin:

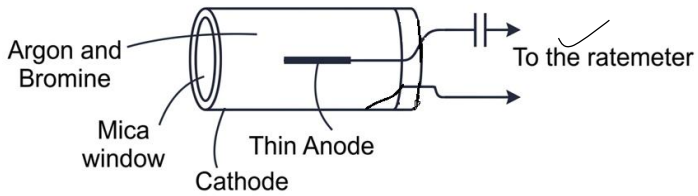


Air above the ball moves with fast speed which creates a low pressure by Bernoulli's principle. The lower surface of the ball moves slower, which creates a high pressure. It experiences a sideways force which causes it to curve the flight. This is due to air being dragged round the spinning ball, thereby increasing the airflow on one side and decreasing it on the other side which causes the pressure difference between the sides

(b) Generation of lift force on aircraft wings:
Aircraft wings are shaped so that air travels faster over the upper surface than the lower surface. By Bernoulli's theorem, a low pressure is developed on the upper surface compared to the lower. The pressure difference produces a lift force on the aircraft.

ITEM 2

Working of a Geiger-Müller Tube (GMT):



– Ionizing radiation enters through the thin mica window.
– This radiation ionizes gas molecules inside the tube.
– Electrons are accelerated to the anode by the E.H.T., and positive charges flow to the cathode.
– A current pulse is produced and it is amplified into the ratemeter, which registers counts per second (activity).
– The magnitude of the count rate recorded determines the type of ionising radiation.
– Alpha particles produce the highest count rate, followed by beta particles, and gamma the least.

Mass of Uranium:

$$A_R = \left(\frac{Area\ of\ window}{Area\ of\ surroundings}\right) \times Activity\ of\ source\ (A_s)$$

$$A_s = (4\pi r^2 \cdot A_R) / S$$
$$= 4\pi(12.5)^2(1.78 \times 10^8) / 250$$
$$A_s = 1.14 \times 10^8\ Bq$$

Decay constant:

$$\lambda = ln\,2 / t½ = ln\,2 / (30 \times 365 \times 24 \times 3600) = 7.33 \times 10^{-10}\ s^{-1}$$

Number of atoms N (from A = λN):

$$N = A_s / \lambda = 1.14 \times 10^8 / 7.33 \times 10^{-10} = 1.05 \times 10^{19}\ atoms$$

Mass of Uranium:

$$M = \frac{N \cdot M_r}{N_A} = (1.05 \times 10^{19} \times 238) / (6.02 \times 10^{23})$$
$$M = 4.151 \times 10^{-3}\ g\ or\ 4.151 \times 10^{-6}\ kg$$

Energy released by 1 atom (nuclear reaction):

Mass of left side (reactants):

$$M_L = 390.19 \times 10^{-27} + 1.67 \times 10^{-27} = 391.86 \times 10^{-27}\ kg$$

Mass of right side (products):

$$M_R = 152.57 \times 10^{-27} + 233.92 \times 10^{-27} + 3(1.67 \times 10^{-27}) = 391.50 \times 10^{-27}\ kg$$

$$\Delta M = 0.36 \times 10^{-27}\ kg$$
$$E = \Delta Mc^2 = 0.36 \times 10^{-27} \times (3 \times 10^8)^2 = 3.24 \times 10^{-11}\ J$$

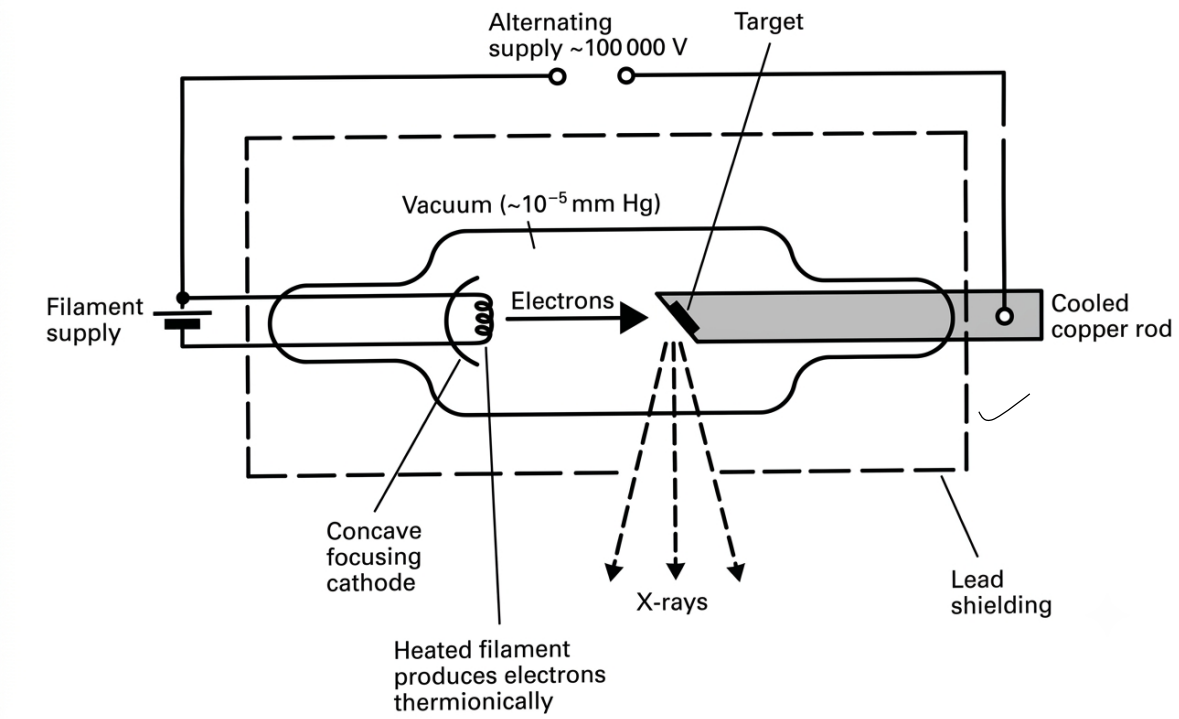
Total energy for all atoms:

$$E_T = N \cdot E = 1.05 \times 10^{19} \times 3.24 \times 10^{-11} = 3.402 \times 10^8\ J$$

How electricity is produced (from the reactor):

The reactor core containing U-235 undergoes fission and produces heat from the fission reaction. Each fission releases energy E = 3.24 × 10^{-11} J, which is used to heat water surrounding the core. The water is heated to boil and produce steam. The steam is driven at high pressure to drive the turbines of the generator to produce electricity.

(b) The suggested device is an X-ray machine.



– The filament is heated by low pd to produce electrons by thermionic emission.
– The produced electrons are accelerated to the anode by E.H.T.
– When electrons strike the metal target, a large percentage of their energy is converted into anode heat, which is conducted away by the cooling system.
– A small percentage of electron energy is converted into X-rays.
– The X-rays are directed only to the body part with suspected fracture with the help of the window (adjustable).
– A photographic plate placed behind the body part is developed and studied.
– The bones appear white while tissue appears dark. Dark parts in the bone structure indicate a fracture.

ITEM 3

(a) Given: P = 60 W

$$Q = nPt = 140(60)(80 \times 60) = 40{,}320{,}000\,J$$

$$Q = mc\Delta\theta, \quad m = \rho V = 1.2 \times 3200 = 3840\,kg$$

$$40{,}320{,}000 = 3840(1050)\Delta\theta$$

$$\Delta\theta = 10°C$$

Since $\Delta\theta > 9°C$, the fan should be installed.

Wave equation given:

$$y = a\,sin(6.6 \times 10^3 t - 20x) \equiv a\,sin(2\pi ft - 2\pi x/\lambda)$$

$$2\pi f = 6.6 \times 10^3$$

$$f = 3.3 \times 10^3 / \pi = 1050\,Hz$$

$$20 = 2\pi/\lambda$$

$$\lambda = \pi/10 = 0.314\,m$$

$$v = \lambda f = (\pi/10)(3.3 \times 10^3/\pi) = 330\, m\, s^{-1}$$

Speed of sound in air (from formula):

$$V_2 = \sqrt{(\gamma P/\rho)} = \sqrt{(1.4 \times 1.2 \times 10^5 / 1.2)} = 374.17\, m\, s^{-1}$$

$$\Delta V = V_2 - V = 374.17 - 330 = 44.2\, m\, s^{-1}$$

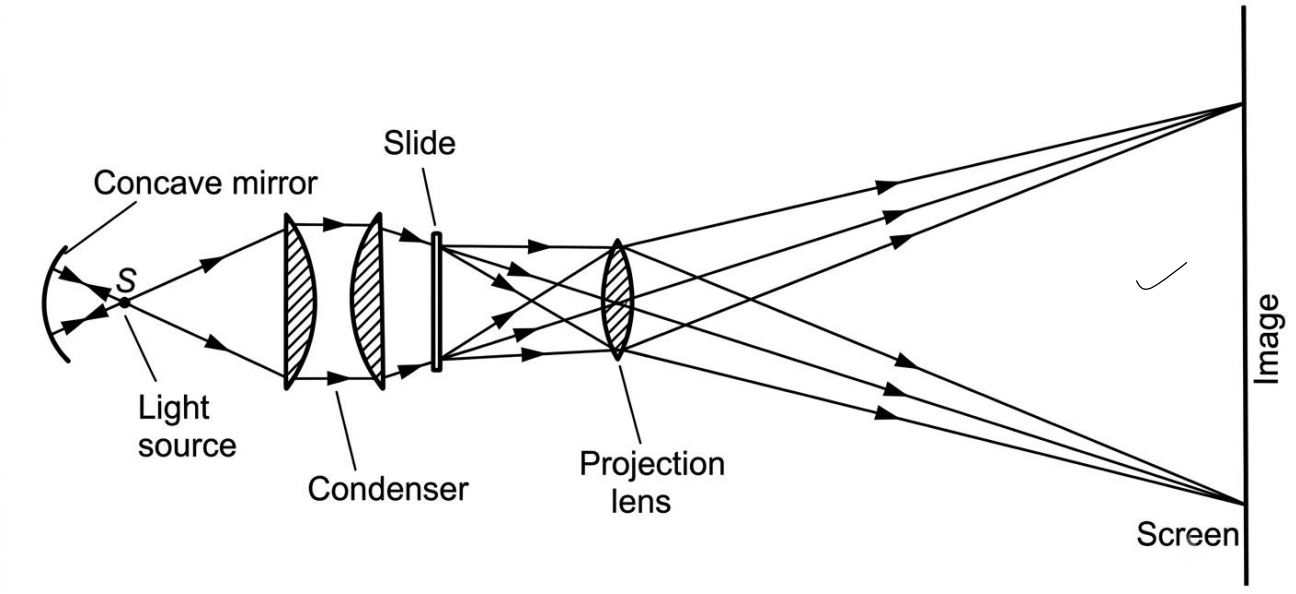
Since 44.2 m s^{-1} < 50 m s^{-1}, this sound poses no hearing challenge.

Justification – Need for Ventilators:
The 10°C temperature rise heats the room, which causes the air to expand (intermolecular spacing increases). The warm, less dense air rises and escapes through ventilators.

Factors affecting sound in the classroom:

– Temperature of the air: Sound speed increases with an increase in temperature. The sound is refracted to bend down from the warmer regions near the ceiling.
– Distance: The intensity of sound reduces as the distance from the source increases.
– Interference (any 2): Reflected sound waves interfere with the original sound creating regions of low sound (destructive interference) and regions of high sound (constructive interference).

(b) The optical device is the projector.



A strong light source is placed at the centre of curvature of a concave mirror which reflects light back and the light is directed to the condenser lens. The condenser lens converges the light through the slide on to the projection lens. The projection lens is adjustable to produce a sharp magnified real image on the screen.

ITEM 4

Angular frequency:

$$\omega = \sqrt{(k/m)} = \sqrt{(504/0.2)} = 50.2\, rad\, s^{-1}$$

Frequency:

$$f = \frac{\omega}{2\pi} = 50.20 / 6.283 = 8.0\, Hz$$

At point P:

$$\lambda = v/f = 1.6 / 8.0 = 0.20\, m$$

$$Path\ difference = S_1P - S_2P = 1.8 - 1.4 = 0.4\, m$$

$$No.\ of\ wavelengths = path\ difference / \lambda = 0.4 / 0.2 = 2.0$$

∴ Constructive interference occurs at P since the wavelength is a whole number multiple of the path difference.

Slightly after P, destructive interference occurs since the path difference may no longer be a multiple of the wavelength.

Heat required to raise water from 20°C to 100°C:

$$Q_water = mc\Delta\theta = 2 \times 4200(100 - 20) = 672{,}000\, J$$

Heat absorbed by kettle:

$$Q_kettle = c\Delta T = 420 \times 80 = 33{,}600\, J$$

$$Total\ heat = 672{,}000 + 33{,}600 = 705{,}600\, J$$

$$Time\ taken = Total\ heat / Power = 705{,}600 / 2352 = 300\, s = 5\, minutes$$

Since the kettle boils the water in 5 minutes, the kettle is genuine.

Energy required to vaporize water:

$$Q = P \times t = 2352 \times 240 = 564{,}480\, J$$

Mass of water vaporized:

$$m = Q / L_v = 564{,}480 / (2.25 \times 10^6) = 0.251\, kg$$

Volume of steam produced:

$$\rho = m/V \implies V = m/\rho = 0.251 / 0.6272 = 0.400\, m^3$$

Volume of water vaporized:

$$V_{water} = m / \rho_{water} = 0.251 / 1000 = 2.51 \times 10^{-4}\, m^3$$

Change in volume:

$$\Delta V = 0.4 - 0.000251 = 0.3997\, m^3$$

Work done against atmospheric pressure:

$$W = P_{atm} \times \Delta V = 1.0 \times 10^5 \times 0.3997 = 39{,}970\, J$$

$$\therefore Work\ done = 39.97\, kJ$$

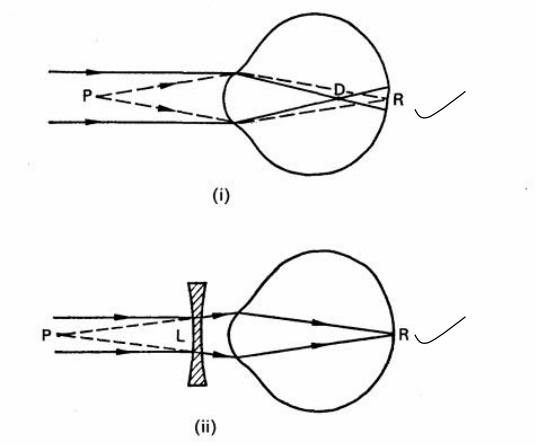
Molecular account of all heating processes:
Initially, heat energy supplied increases the average kinetic energy of water molecules. Intermolecular distances remain almost constant, but molecules vibrate vigorously within the liquid structure. Since temperature is proportional to kinetic energy, temperature increases but no change in state occurs as molecules remain bonded in liquid clusters.

When the liquid reaches boiling point (100°C), temperature remains constant since all energy supplied is used to overcome intermolecular forces of attraction. The molecules gain enough energy to break free from the liquid surface and expand as vapour. Steam molecules move rapidly with large separations since intermolecular forces are negligible.

The work done by steam (39.97 kJ) is used in pushing back the atmosphere as the steam expands.

(b) Recommended device for Rose: Concave lens

How it helps correct short sightedness:



Short sightedness occurs when the focal length of the eye is too short, owing to the eyeball being too long. Parallel rays will be brought to a focus at point D in front of the retina.

It is corrected by using diverging lenses. Parallel rays refracted through L are now made divergent, and if they appear to come from the far point P of the eye, the rays are brought to a focus on the retina R.

ITEM 5

(a) $E = I(R + r)$

$E = 0.48(21 + r)$……(i)

$E = 0.30(36 + r)$……...(ii)

(i) = (ii)

$0.48(21 + r) = 0.30(36 + r)$

$r = 28\Omega$

The internal resistance of the battery is greater than 20Ω which makes it not good for charging

$E = 0.30(36 + 28) = 19.2$

$$c = \frac{c_1 c_2}{c_1 + c_2} = \frac{1 \times 1}{1 + 1} = 0.5F$$

Charging the capacitor by a battery

When the capacitor is connected to the battery, electrons flow from the negative terminal of the battery to the plates of the plate of the capacitor making the plate to accumulate the negative charges while removing them from the other plate to make it positive

An electric field is created between the plates which stores energy. The charging continues until the voltage on the capacitor plates is equal to the voltage of the battery

$$energy = \frac{1}{2}CV^2 = \frac{1}{2} \times 0.5 \times 19.2^2 = 92.16J$$

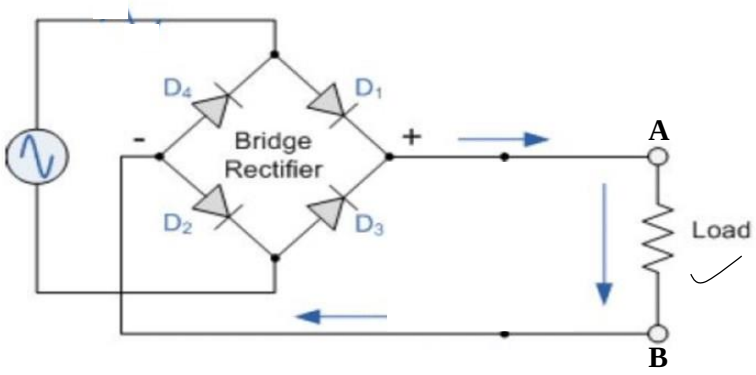
$$power = I^2(R + r) = 0.3^2(36 + 28) = 5.76W$$

$$power \times time = energy$$

$$time = \frac{energy}{power} = \frac{92.16}{5.76} = 16s$$

They used to charge for only 15 seconds thus the capacitor was never fully charged

(b) This is done by using diodes D_1, D_2, D_3 and D_4 connected in a bridge circuit as shown below



– During the positive half cycle of A.C, D_1 and D_2 conduct; current flows through the load from A to B.

– During the negative half cycle, D_3 and D_4 conduct; current still flows through the load from A to B.

– Current always flows through the same direction through the load and it flows in both cycles.

ITEM 6

(a) $C = \frac{\varepsilon_r \varepsilon_0 A}{d} = \frac{4 \times 8.85 \times 10^{-12} \times 0.0005}{0.02} = 8.85 \times 10^{-13} F$

Total capacitance $= \frac{c_1 c_2}{c_1 + c_2} = \frac{8.85 \times 10^{-13} \times 8.85 \times 10^{-13}}{8.85 \times 10^{-13} + 8.85 \times 10^{-13}} = 4.425 \times 10^{-13} F$

Inductance, $L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi \times 10^{-7} \times 10^2 \times 0.0012}{1.0 \times 10^{-2}} = 1.507 \times 10^{-6} H$

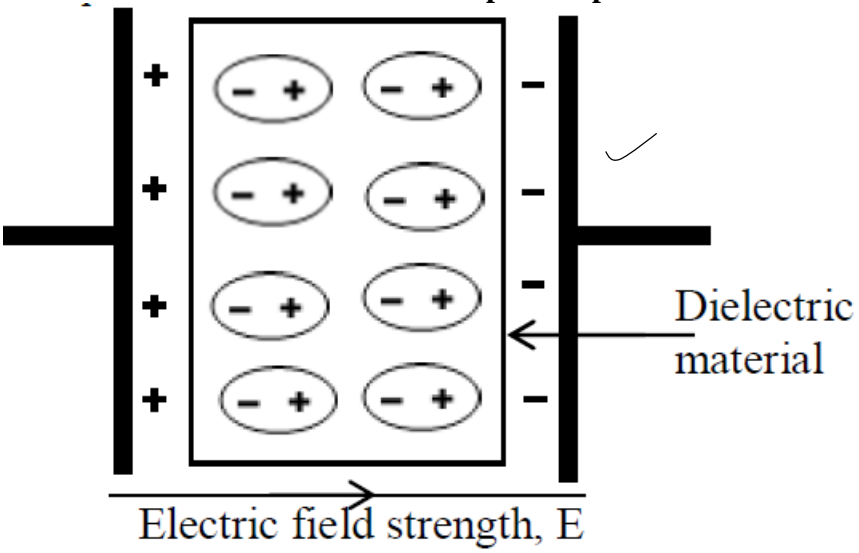
Resonant frequency $f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1.507 \times 10^{-6} \times 4.425 \times 10^{-13}}} = 194.9 \times 10^6 Hz$

Since the value of resonant frequency (194.9MHz) exceeds the required 61.2MHz of the channel, hence the new circuit will tune to the required frequency.

The exact capacitor should be of capacitance given by

$C = \frac{1}{4\pi^2 f^2 L} = \frac{1}{4\pi^2 \times (61.2 \times 10^6)^2 \times 1.507 \times 10^{-6}} = 4.488 \times 10^{-12} F$ which is required to tune to the exact resonant frequency of the religious channel

Effect of a dielectric between the capacitor plates



- When the dielectric is sandwiched between the plates of an isolated charged capacitor, the molecules of the dielectric get polarized by the electric field intensity between the plates.
- At the surface of the dielectric, charges appear of opposite sign to those of the nearest (closest) plate. But since these are bound charges,, an electric field develops between the opposite faces of the dielectric in a direction opposite to the applied field.
- This reduces the resultant electric field intensity and hence the P.d between the plates also reduces (since $= \frac{V}{d}$).
- Since $C = \frac{Q}{V}$, it implies that the capacitance, C , increases.

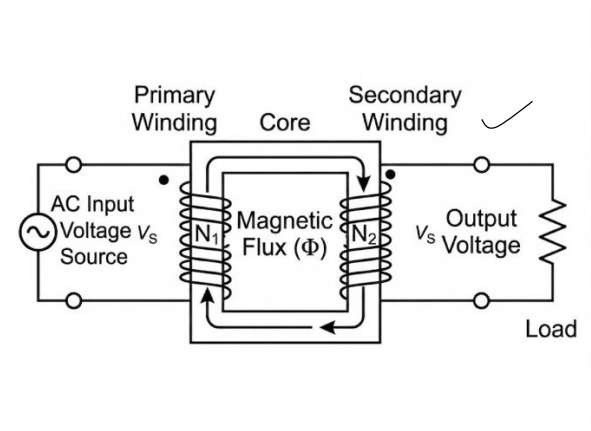
(b) **The device is a step down transformer(a transformer)**

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

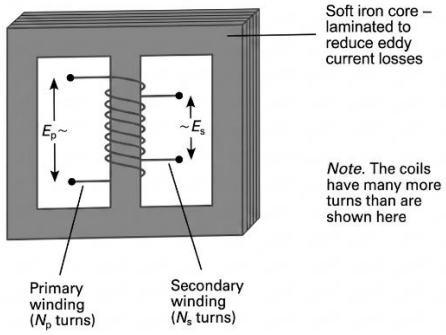
$$\frac{240}{12} = \frac{N_p}{N_s}$$

$$N_p = 20N_s$$

The charger has a transformer whose number of turns in the primary coil must be 20 times bigger than the turns in the secondary coil.



Alt:



An alternating voltage in the primary coil where the charger is inserted in the socket produces alternating current in the coil.

This current produces a changing magnetic flux in the soft iron core which links up the secondary coil of fewer turns. An alternating emf (12V) is induced in the secondary coil which is used to charge the phone

END